Yinzhe Zhou

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EDUCATION

Johns Hopkins University — Baltimore, MD, USA
Master of Science in Electrical and Computer Engineering — *Aug. 2025 – Present*

- **GPA:** 3.73/4.0
- **Relevant Coursework:** FPGA Synthesis Lab, CAD Design of Digital VLSI Systems, Advanced Topics in Microsystem Fabrication, Neural Implants and Interfaces, Mixed-Mode VLSI Systems.

University of Macau — Macao SAR
Bachelor of Science in Electrical and Computer Engineering — *Aug. 2021 – May 2025*

City University of Hong Kong — Hong Kong SAR
Exchange Student, Department of Electrical Engineering — *Jan. 2023 – Aug. 2023*

PUBLICATIONS

HACo: Learning Haptic Active Compliance for Force-Aware Dexterous Manipulation
Naisheng Ye, **Yinzhe Zhou**, Junkai Zhao, Yuhang Lu, Checheng Yu, Zhenjie Yang, Pengwei Wang, and Hongyang Li. *arXiv:2609.36596, 2026.*

OpenTac: An Open Ecosystem for Customizable, Scalable, and Transferable 3D Tactile Sensing
Ariel Slepyan, Geeta Chandra Raju Bethala, Jiafeng Gu, Anagha Rajendra Dangle, **Yinzhe Zhou**, Hongyuan Fu, Jade Sim, Tuhin Basu, Xinyi Yin, Md Faizal Karim, Gregory Hager, Homanga Bharadhwaj, Anand Bhattad, Krishna Murthy Jatavallabhula, and Nitish V. Thakor. *Manuscript, 2026.*

INTERNSHIPS

Beijing Academy of Artificial Intelligence (BAAI) — Beijing, China
Research Intern — *Jun. 2026 – Sep. 2026*

HACo: Haptic Active Compliance for Dexterous Manipulation

- Developed compliance-regulated teleoperation for dual UR5 arms with Sharpa dexterous hands, combining Cartesian admittance and fingertip force regulation; collected **500 demonstrations** across five tasks with synchronized vision, robot state, tactile, and joint-torque observations.
- Developed a kinematics-aligned haptic encoder fusing ten fingertip tactile streams with 44 hand-joint torques, and gated cross-attention to condition a pretrained vision-language-action model on contact and load feedback.
- Trained the policy with conditional flow matching on compliant-action targets and auxiliary compliant-intent supervision; integrated real-time action chunking for asynchronous closed-loop execution.
- Conducted real-robot comparisons and component ablations across **five force-sensitive tasks**, with 20 trials per task–method pair; HACo achieved **83% mean success**, versus **35%** for the strongest evaluated baseline (**+48 percentage points**).

Microsoft Research Asia (MSRA) — Beijing, China
Research Intern — *Jun. 2024 – Sep. 2024*

Fine-Tuning Mixtral 8x7B on Biochemical Molecular Data

- Constructed and preprocessed a PubChem-based dataset for molecular property prediction tasks.
- Implemented KNN retrieval over SMILES representations to provide similar-molecule context for in-context learning.
- Fine-tuned Mixtral 8x7B and refined the evaluation protocol, improving test accuracy from **67% to 77%**.
- **Tech:** Python, Git, VS Code.

RESEARCH EXPERIENCE

Johns Hopkins University × TU Darmstadt — Baltimore, MD, USA
Graduate Researcher — *Nov. 2025 – Present*

Event-Based Visuo-Tactile Sensing for Robot Imitation Learning

- Building a multimodal imitation-learning pipeline that fuses RGB, tactile, and event streams for a UR robot-mounted gripper, targeting contact, slip, and fast-motion cues missed by RGB-only policies.
- Implemented a synchronized multi-rate data pipeline from handheld demonstrations into observation–action samples, aligning RGB at 30 Hz, tactile/event streams at 120 Hz, and 3D pose at 75 Hz.
- Trained and deployed **Diffusion Policy** and **Flow Matching** algorithms on real-world tasks.
- Developing an **event-conditioned Flow Matching** policy with a lightweight 3D-CNN event encoder for high-frequency tactile/event observations.

Johns Hopkins University Baltimore, MD, USA
Graduate Researcher *Feb. 2026 – Sep. 2026*

OpenTac: Shear-Force Simulation and Robot Experiments

- Developed shear-force sensing simulations in Isaac Sim for robot contact scenarios.
- Contributed to integrating scalable piezoresistive tactile skins onto a UMI handheld gripper for robot data collection.
- Designed and ran sensor characterization and robotic grasping experiments to evaluate normal- and shear-force feedback under varied contact conditions.

University of Macau Macao SAR
Final Year Project — Faculty Dean's Honors List *Sep. 2024 – May 2025*

Singular Spectrum Analysis for Biosignal Denoising and Feature Extraction

- Designed a two-level Singular Spectrum Analysis framework combining component correlation and waveform reconstruction for ECG and EEG signal analysis.
- Localized ECG PQRST features and removed EEG artifacts while preserving neural rhythms across delta, theta, alpha, and beta bands.
- Validated the method on synthetic and real recordings, recovering physiological patterns such as eyes-open vs. eyes-closed alpha rhythm differences.
- **Tech:** Python, NumPy, SciPy, Matplotlib, NeuroKit2.

City University of Hong Kong Hong Kong SAR
Research Assistant *May 2023 – Aug. 2023*

Neural Decoding for Implantable Brain–Machine Interfaces with ANN and SNN

- Preprocessed raw primate neural recordings into supervised datasets with movement coordinates, finger positions, and spike trains.
- Designed and benchmarked ANN and SNN decoders for neural decoding, comparing accuracy–efficiency trade-offs for implantable BMI scenarios.
- **Tech:** Python, MATLAB.

University of Macau Macao SAR
Research Assistant *May 2022 – Jul. 2022*

FPGA-Based Hardware Time-Triggered System for Biomedical Engineering

- Implemented FPGA-based signal conditioning and filtering pipelines in Quartus for biomedical data acquisition.
- Refined hardware architecture and I/O interfaces to support deterministic time-triggered behavior.
- **Tech:** Quartus, MATLAB.

TEACHING EXPERIENCE

Johns Hopkins University Baltimore, MD, USA
Course Assistant, Hands-on Robot Learning (EN.601.498.01.FA26) *Aug. 31 – Dec. 23, 2026*
Instructor: Krishna Murthy Jatavallabhula.

SELECTED HONORS & AWARDS

1st Runner-Up, IEEE Student Branch Project Competition 2025.

Akrostar Academic Prize, University of Macau, 2024/2025.

Dean's Honor List, University of Macau, all semesters.

Academic Excellence Outgoing Exchange Scholarship, City University of Hong Kong, 2023.

Top 10%, Huawei ICT Competition (Preliminary), Macau, 2021–2022.

TECHNICAL SKILLS

Programming: Python, C/C++, MATLAB, Verilog, VHDL.

Robotics / ML: PyTorch, ROS 2, Isaac Sim, Diffusion Policy, Flow Matching.

Sensing / Hardware: Embedded firmware, data acquisition, FPGA development, Vivado, Quartus, Cadence.